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Nonlinear plasma physics: from waves to observable signals

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How does a coherent signal survive in a turbulent environment? I study this question through fast radio bursts, combining nonlinear theory and large-scale simulations of electromagnetic energy around neutron stars and black holes. My work involves algorithm design, terabyte-scale datasets, and simulations using 10^3–10^4 MPI processes. In June 2026, I transferred from the University of Maryland to Stanford to continue my Ph.D.

Can a radio burst escape its source?

Fast radio bursts (FRBs) are exceptionally bright radio flashes lasting microseconds to milliseconds. Magnetars, neutron stars with enormous magnetic fields, are promising sources. Yet producing a burst is only half the problem: the signal must also cross the surrounding plasma without losing the properties we observe. This makes propagation a powerful test of emission models.

In a magnetically dominated plasma, a radio signal can behave as a fast magnetosonic wave. Nonlinear interactions can transfer its energy into Alfvén waves, disturbances guided along magnetic field lines. A useful mathematical description starts with resonant interactions among three modes:

$$\mathbf{k} = \mathbf{p} + \mathbf{q}, \qquad \omega_{\mathbf{k}} = \omega_{\mathbf{p}} + \omega_{\mathbf{q}}.$$

These conditions express matching of wavevectors and frequencies. The difficulty is that one mode can interact with many resonant pairs, and energy transfer need not proceed through neighboring length scales. Analytical estimates therefore need to be tested against the nonlinear evolution of the full system.

My first 2026 preprint [1] uses force-free electrodynamics simulations, which neglect plasma inertia in the strongly magnetized limit, to study this evolution. We find that fast magnetosonic waves efficiently generate secondary fast and Alfvén waves. Energy continues to leave the original wave even after the two components carry comparable energies, while the excited spectrum broadens. The result connects a tractable theoretical instability to the eventual loss of a coherent signal.

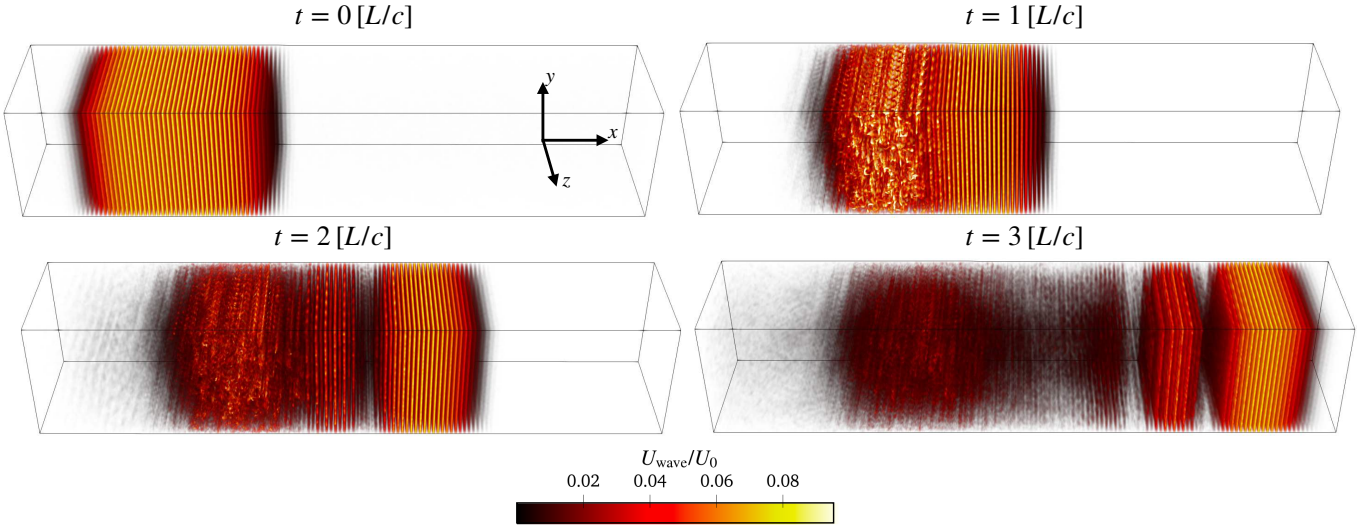


Figure 1: A pulse propagating to the right in a force-free simulation develops small-scale structure. Colors show normalized wave energy density; time is measured in units of L/c, with L the box length and c the speed of light. The sequence illustrates how nonlinear interactions reprocess an initially coherent wave.

From an instability to constraints on fast radio bursts

In a companion 2026 preprint [2], we apply this mechanism to radio-burst propagation through magnetar magnetospheres. Three-dimensional simulations show strong attenuation as the signal excites Alfvénic fluctuations. The physical limit is as important as the instability itself: the plasma must supply the electric current required by those fluctuations. If it cannot, "charge starvation" can interrupt the decay.

Our analysis finds efficient decay within roughly 10–100 stellar radii in a quiescent dipolar magnetosphere. In relativistic magnetic outflows launched by eruptions, the interactions can remain effective farther out, requiring luminous bursts to escape from radii of order 10^2–10^3 stellar radii or larger in the scenarios studied. These results constrain the conditions under which inner-magnetospheric models can work; they do not establish that every such model is impossible. This distinction motivates my next step: identifying where the simplified description breaks down and which kinetic processes determine the escaping signal.

Next direction: resolving the plasma response

I propose to use particle-in-cell (PIC) simulations, which evolve charged particles and electromagnetic fields together, to examine the transition from nonlinear wave decay to current-limited behavior. This approach can capture particle heating and radiative cooling that force-free simulations omit. My central question is whether these processes suppress the instability, redirect its energy, or produce additional observable signatures.

I would begin with controlled problems that overlap with the force-free regime, comparing growth rates, energy budgets, and wave spectra. I would then vary plasma density, wave amplitude, and cooling strength to locate the boundary between strong reprocessing and transmission. Resolution and domain-size studies would distinguish physical damping from numerical dissipation. The goal is to map where a pulse survives, broadens, or deposits energy into particles, with explicit uncertainties.

This direction builds on my participation in the development of Entity, an open-source, hardware-portable PIC code [3]. The project uses C++ and Kokkos to support different CPU and GPU architectures. My experience with MPI, OpenMP, code optimization, and large-data analysis helps me connect the physics question to a calculation that is feasible to run and possible to validate.

A broader computational research program

In my first-author study of M87* [4], we combined general relativistic ray tracing with a simulated reconnecting current sheet to model very-high-energy flares. Accounting for lensing and directional emission links the evolving plasma to the observed light curve. Across these projects, I aim to build models whose assumptions can be tested from the governing equations to the final observable.

The fellowship would support kinetic tests of the range of validity of my force-free results. I would bring experience with nonlinear systems, parallel computation, and extracting reliable conclusions from complex simulations.

Selected references

[1] S. Solanki, J. Mahlmann, A. Philippov. *Nonlinear Decay of Fast Magnetosonic Waves through Weak Turbulence: Force-Free Electrodynamics Simulations.* 2026. arXiv:2606.19450.

[2] S. Solanki, J. Mahlmann, A. Philippov, A. Beloborodov. *Damping of Fast Radio Bursts in the Inner Magnetospheres of Magnetars.* 2026. arXiv:2606.19448.

[3] H. Hakobyan et al. (including S. Solanki). *Entity – Hardware-agnostic Particle-in-Cell Code for Plasma Astrophysics. I.* ApJS, 285, 11 (2026). arXiv:2511.17710.

[4] S. Solanki et al. *Modeling of Lightcurves from Reconnection-powered Very High-energy Flares from M87*.* ApJ, 985, 147 (2025). arXiv:2504.16298.